Reasoning Trace Length as a Proxy for Robustness: A Systematic Investigation of Length–Accuracy Relationships Across In-Distribution and Out-of-Distribution Tasks

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Abstract

Chain-of-thought reasoning is the dominant paradigm for improving large language model (LLM) performance, yet practitioners lack guidance on how reasoning trace length interacts with out-of-distribution (OOD) robustness. We systematically investigate whether longer reasoning chains always improve generalization by varying token budgets across five levels in GPT-4.1 and evaluating on four benchmarks: MATH (in-distribution), GSM8K (easy OOD), MMLU-Pro (knowledge-heavy OOD), and MuSR (hard multi-step reasoning OOD). We find that the length–accuracy relationship is *task-dependent*: monotonically increasing for tasks within the model's capability range (MATH: 46%→78%; MMLU-Pro: 63%→79%), flat for easy tasks (GSM8K: ∼93% across all budgets), and non-monotonic for tasks at the model's capability frontier (MuSR peaks at 22% with moderate budgets before declining). Critically, we discover that model-expressed confidence is well-calibrated in-distribution (ρ=0.63 on MATH) but completely uncalibrated on hard OOD tasks ($\rho\approx0$ on MuSR), undermining confidence-based adaptive length selection precisely where it is most needed. An adaptive two-pass strategy achieves comparable accuracy to fixed long reasoning on MATH with 32% fewer tokens, but fails catastrophically on MuSR due to systematic overconfidence. These findings reveal that the "more reasoning is better" assumption breaks down at the frontier of model capability, with direct implications for efficient and robust LLM deployment.

1 Introduction

Chain-of-thought (CoT) reasoning has become the default strategy for improving LLM performance on complex tasks [Wei et al., 2022]. Large reasoning models such as OpenAI's o1 [OpenAI, 2024] and DeepSeek-R1 [DeepSeek-AI, 2025] generate extended internal reasoning traces spanning thousands of tokens. The prevailing assumption is straightforward: more reasoning leads to better answers. But does this assumption hold when models encounter tasks outside their training distribution?

Recent work challenges the "more is always better" narrative. Wu et al. [2025] establish an inverted U-shaped curve between CoT length and accuracy on in-distribution (ID) tasks, showing that excessively long traces accumulate errors. Su et al. [2025] demonstrate that incorrect responses are dramatically longer than correct ones, while Yang et al. [2025] reveal that training-free length reduction degrades behavioral consistency even when accuracy is preserved. These findings raise a natural question: *how does the length–accuracy relationship change when models face out-of-distribution tasks, and can we exploit this relationship for more robust deployment?*

Despite growing interest in efficient reasoning [Sui et al., 2025, Yue et al., 2025], no prior work has systematically measured how the reasoning length–accuracy curve shifts across ID versus OOD

benchmarks, or whether uncertainty signals can guide adaptive length selection for OOD robustness. Aggarwal and Welleck [2025] show that RL-trained length control generalizes to some OOD tasks, but do not characterize the shape of the length–robustness curve. Valmeekam and Kambhampati [2025] find that semantically corrupted traces can match or exceed genuine traces on OOD tasks, suggesting that trace content and length may be decoupled from actual reasoning quality.

We address this gap with a systematic investigation of reasoning trace length as a proxy for robustness. Using budget-forcing prompts, we control reasoning length at five levels (from zero reasoning to unconstrained) in GPT-4.1 and evaluate on four benchmarks spanning different domains and difficulty levels: MATH [Hendrycks et al., 2021] (ID, competition math), GSM8K [Cobbe et al., 2021] (easy OOD), MMLU-Pro [Wang et al., 2024] (knowledge-heavy OOD), and MuSR [Sprague et al., 2023] (hard multi-step reasoning OOD). We measure accuracy, token usage, confidence calibration, and the *robustness gap*—the ID–OOD accuracy difference—as a function of trace length.

Our results reveal a nuanced picture. On MATH and MMLU-Pro, accuracy increases monotonically with trace length (46%$\rightarrow$78% and 63%$\rightarrow$79%, respectively), with diminishing returns beyond $\sim$500 tokens. On GSM8K, accuracy is flat at $\sim$93% regardless of budget. On MuSR, however, we observe the hypothesized non-monotonic pattern: short reasoning (5.1%) is *worse* than no reasoning (10.3%), accuracy peaks at moderate length (21.8%), and declines slightly at the longest budget (20.5%). Most strikingly, model confidence is well-calibrated ID (ρ=0.63 on MATH) but completely uninformative OOD ($\rho\approx0$ on MuSR), causing a confidence-based adaptive strategy to fail precisely where it is most needed.

We make the following contributions:

- We conduct the first systematic study of how the reasoning trace length–accuracy relationship varies across ID and OOD benchmarks, revealing that the relationship is task-dependent rather than universally non-monotonic.
- We identify the "underthinking trap"—a novel finding that forcing brief reasoning can be *worse* than no reasoning at all on hard OOD tasks, with short CoT underperforming no-CoT by 5.2 percentage points on MuSR.
- We demonstrate that model-expressed confidence is catastrophically miscalibrated on hard OOD tasks ($\rho\approx0$), undermining confidence-based adaptive length selection and revealing a fundamental limitation of self-supervised uncertainty estimation for reasoning control.
- We evaluate an adaptive two-pass strategy that achieves 32% token savings on ID tasks while matching fixed-budget accuracy, but fails on OOD tasks due to overconfidence—highlighting the need for external calibration signals.

2 Related Work

Chain-of-thought reasoning and length effects. Wei et al. [2022] introduced chain-of-thought prompting, demonstrating that intermediate reasoning steps substantially improve LLM performance on arithmetic and commonsense tasks. Subsequent work revealed that the relationship between reasoning length and quality is more complex than initially assumed. Wu et al. [2025] provide the most comprehensive analysis to date, establishing an inverted U-shaped curve between CoT length and accuracy using both empirical evaluation and a formal probabilistic model based on the Lambert W function. They show that the optimal CoT length increases with task difficulty but *decreases* with model capability—a "simplicity bias" where stronger models benefit from more concise reasoning. Su et al. [2025] complement this by characterizing two failure modes at opposite extremes: *underthinking* (insufficient reasoning on hard problems) and *overthinking* (excessive reasoning on easy problems). Our work extends these findings by examining how the length–accuracy curve shifts across ID and OOD domains, a dimension not explored in prior studies.

Efficient reasoning and length control. A growing body of work addresses the computational cost of extended reasoning. Aggarwal and Welleck [2025] propose Length-Controlled Policy Optimization (LCPO), an RL-based approach that enables precise user-specified token budgets while maintaining accuracy. Their method achieves log-linear scaling of performance with reasoning length and demonstrates some OOD generalization. Chen et al. [2025] take a training-free approach via steering vectors, categorizing CoT into execution, reflection, and transition thoughts, and showing that excessive reflection correlates with failure. Surveys by Sui et al. [2025] and Yue et al. [2025] organize the rapidly growing literature into model-based, output-based, and prompt-based efficiency

methods. Unlike these approaches that focus primarily on reducing length while preserving ID accuracy, we investigate how length control affects OOD robustness—a distinct and complementary question.

Reasoning quality beyond length. Several papers argue that trace structure, not just length, determines reasoning quality. Jiang et al. [2025] convert sequential reasoning traces into hierarchical trees and show that structural features (branching patterns, backtracking frequency, verification steps) predict correctness better than length alone, achieving +5.63% improvement in tree-based Best-of-N voting. Bogdan et al. [2025] introduce "thought anchors"—individual sentences with outsized counterfactual impact on final answers—suggesting that reasoning traces contain both critical and redundant components. Perhaps most provocatively, Valmeekam and Kambhampati [2025] demonstrate that models trained on *semantically corrupted* reasoning traces achieve comparable and sometimes *better* OOD generalization than those trained on correct traces, challenging the assumption that trace content reflects genuine reasoning. Massoli et al. [2026] provide a theoretical framework by recasting efficient reasoning as lossy compression under the Information Bottleneck principle.

Robustness and behavioral consistency. Yang et al. [2025] expose hidden costs of training-free length reduction: even when accuracy is preserved, behavioral consistency degrades substantially. Their ICBench benchmark reveals that budget-forcing and no-thinking strategies nearly double inconsistency scores and increase sycophancy, with models concealing their decision-making process in shortened outputs. Dellibarda Varela et al. [2025] show that token usage reflects a model's *self-assessed tractability* rather than actual reasoning quality—when models "give up" on hard problems, token usage drops sharply. These findings motivate our investigation of confidence calibration as a function of trace length and task difficulty: if models are unreliable narrators of their own reasoning process, confidence-based adaptive strategies may fail in predictable ways.

Position of our work. Prior work has established that (i) the length–accuracy relationship is non-monotonic on ID tasks [Wu et al., 2025], (ii) length reduction can degrade behavioral consistency [Yang et al., 2025], and (iii) RL-trained length control can generalize to some OOD settings [Aggarwal and Welleck, 2025]. However, no prior study has systematically characterized how the full length–accuracy curve shifts across multiple OOD domains, measured the *robustness gap* as a function of trace length, or evaluated whether model confidence can guide adaptive length selection across the ID–OOD spectrum. Our work fills this gap.

3 Methodology

3.1 Overview

We study the relationship between reasoning trace length and accuracy across four benchmarks by controlling token budgets via system-prompt instructions. Our experimental design has three components: (1) budget-forcing at five levels to vary reasoning length, (2) evaluation on benchmarks spanning different domains and difficulties, and (3) an adaptive two-pass strategy that uses model confidence to select reasoning depth. We use GPT-4.1 as our target model, as it represents a state-of-the-art general-purpose LLM (released April 2025) with strong reasoning capabilities.

3.2 Budget-Forcing Protocol

We control reasoning trace length through system-prompt instructions paired with maximum token limits, following the budget-forcing approach of Aggarwal and Welleck [2025]. We define five conditions spanning the full spectrum from zero reasoning to unconstrained generation:

All queries use temperature $T{=}0.0$ for deterministic outputs and a fixed random seed of 42 for reproducibility. We record the full response text, extracted answer, actual completion token count, and model-expressed confidence (extracted via a follow-up prompt) for each query.

3.3 Adaptive Two-Pass Strategy

We evaluate a simple confidence-based adaptive strategy: (1) query with the SHORT budget; (2) extract the model's self-reported confidence; (3) if confidence $< 75\%$, re-query with the LONG

Table 1: Budget-forcing conditions. Each condition combines a system prompt instruction with a maximum completion token limit. NO-COT suppresses reasoning entirely; UNCONSTRAINED provides generous headroom for extended traces.

Condition	System Prompt Instruction	Max Tokens
NO-COT	"Answer directly with NO reasoning or explanation"	150
SHORT	"Think briefly in 1–2 sentences"	300
MEDIUM	"Think step by step in moderate detail (3–5 steps)"	800
LONG	"Think very carefully and thoroughly step by step"	1500
UNCONSTRAINED	"Think step by step, taking as much space as you need"	3000

Table 2: Benchmark characteristics. We sample 80 questions per dataset (78 for MUSR due to balancing across subtasks). Each benchmark serves a distinct role in our experimental design.

Dataset	Task Type	n	Role	Answer Format
MATH	Competition math	80	In-distribution	Open-ended (\boxed{})
GSM8K	Grade school math	80	Easy OOD	Numeric
MMLU-PRO	Multi-domain knowledge	80	Knowledge OOD	Multiple choice (A–J)
MUSR	Multi-step soft reasoning	78	Hard OOD	Multiple choice

budget and use that answer instead. The threshold of 75% was chosen to represent moderate uncertainty. This tests whether model confidence can serve as a practical signal for allocating additional reasoning effort.

3.4 Benchmarks

We evaluate on four benchmarks chosen to span a range of domains, difficulty levels, and reasoning types:

MATH [Hendrycks et al., 2021] contains competition mathematics problems across five difficulty levels. We use stratified sampling (16 problems per level) to ensure coverage of the difficulty spectrum. This serves as our primary in-distribution benchmark because GPT-4.1 is extensively trained on mathematical reasoning.

GSM8K [Cobbe et al., 2021] contains grade-school arithmetic word problems. As an "easy OOD" benchmark, it tests whether reasoning length matters when tasks are well within the model's capability.

MMLU-Pro [Wang et al., 2024] extends the original MMLU with harder, 10-option multiple choice questions spanning science, law, engineering, and other domains. It serves as a knowledge-heavy OOD benchmark requiring domain expertise rather than pure reasoning.

MuSR [Sprague et al., 2023] presents long narrative passages (500–1000 words) requiring multi-step deductive reasoning across murder mysteries, object placements, and team allocation scenarios. We balance our sample across all three subtasks. This represents our hardest OOD benchmark, testing reasoning under significant distribution shift.

3.5 Evaluation Metrics

Accuracy. We use exact-match accuracy with task-specific normalization: LaTeX expression matching for MATH, numeric comparison (within tolerance) for GSM8K, and letter matching for MMLU-PRO and MUSR.

Robustness gap. We define the *robustness gap* as $\Delta_{\text{rob}} = \text{Acc}_{\text{MATH}} - \text{Acc}_{\text{OOD}}$ for each OOD benchmark at each budget level. A smaller gap indicates more consistent performance across domains.

Token efficiency. We compute efficiency as $\eta = \text{Acc}/\bar{t} \times 100$, where $\bar{t}$ is the mean completion token count, capturing accuracy per unit of computational effort.

Table 3: Accuracy (%) by dataset and budget condition with 95% bootstrap confidence intervals. Best fixed-budget result per dataset in **bold**. The adaptive strategy uses confidence-based escalation from SHORT to LONG.

Dataset	NO-COT	SHORT	MEDIUM	LONG	UNCONSTRAINED	ADAPTIVE
MATH	46.2 [35–58]	62.5 [51–73]	70.0 [59–79]	71.2 [61–81]	**77.5** [69–86]	71.2
GSM8K	92.5 [86–98]	**93.8** [88–99]	90.0 [84–96]	91.2 [85–96]	**93.8** [88–99]	91.2
MMLU-PRO	62.5 [51–73]	72.5 [63–83]	75.0 [65–84]	77.5 [68–86]	**78.8** [69–88]	71.2
MUSR	10.3 [4–18]	5.1 [1–10]	14.1 [6–22]	**21.8** [13–31]	20.5 [13–30]	7.7

Confidence calibration. We measure the Spearman rank correlation (ρ) between model-expressed confidence and binary correctness, separately for each dataset. High ρ indicates the model can reliably distinguish questions it will answer correctly from those it will not.

Statistical analysis. All accuracy estimates are accompanied by 95% bootstrap confidence intervals computed from 1,000 resamples. We use Spearman correlations for monotonicity tests and paired comparisons between budget conditions. We report p-values without multiple-comparison correction where individual tests are of independent interest, and note significance at $\alpha = 0.05$.

4 Results

4.1 Accuracy Across Budget Conditions

Table 3 presents accuracy across all dataset–budget combinations. The length–accuracy relationship varies dramatically by task.

MATH and MMLU-Pro: monotonically increasing. On both MATH and MMLU-PRO, accuracy increases with each budget level. For MATH, accuracy rises from 46.2% (NO-COT) to 77.5% (UNCONSTRAINED), a 31.3 percentage point gain. The jump from NO-COT to MEDIUM is statistically significant (p=0.002), while the gain from MEDIUM to UNCONSTRAINED is not (p=0.28), indicating diminishing returns. MMLU-PRO follows a similar pattern (62.5%→78.8%).

GSM8K: flat. On GSM8K, accuracy is near-ceiling across all conditions ($90-93.8\%$), with no statistically significant differences between any pair of budgets. This confirms that easy tasks within the model's capability show no benefit from extended reasoning.

MuSR: non-monotonic. On MUSR, we observe the hypothesized non-monotonic pattern. SHORT (5.1%) is *worse* than NO-COT (10.3%), suggesting that forcing brief reasoning can be actively harmful. Accuracy then climbs to peak at LONG (21.8%) before declining slightly at UNCONSTRAINED (20.5%). This is the only dataset exhibiting the inverted-U shape.

Figure 1a visualizes these accuracy curves across all budget conditions, and figure 1b shows accuracy as a function of actual token usage.

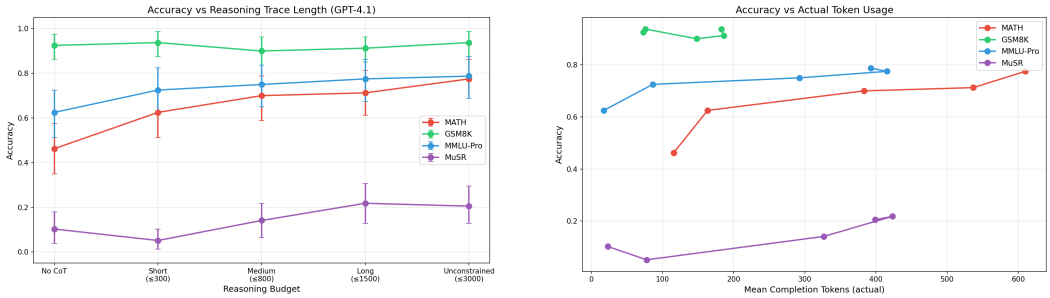
4.2 Token Usage and Self-Regulation

Table 4 reports mean actual token counts. A notable finding is that the model *self-regulates* its output length: the LONG (max 1500) and UNCONSTRAINED (max 3000) conditions produce similar actual token counts ($\sim 400-600$ tokens), suggesting the model reaches a natural stopping point regardless of the allocated budget ceiling.

On GSM8K, the model uses very few tokens even when given unconstrained budgets (~ 183 tokens at UNCONSTRAINED), reflecting its ease with the task. In contrast, MATH elicits longer traces (~ 610 tokens at UNCONSTRAINED), consistent with the higher difficulty.

4.3 Robustness Gap Analysis

Figure 2 shows the *robustness gap* ($\Delta_{\text{rob}} = \text{Acc}_{\text{MATH}} - \text{Acc}_{\text{OOD}}$) across budget conditions. The behavior differs by OOD benchmark:



(a) Accuracy vs. budget condition

(b) Accuracy vs. actual token count

Figure 1: Length–accuracy curves by dataset. *(Left)*Accuracy across five budget conditions. MATH and MMLU-PRO increase monotonically; GSM8K is flat; MUSR shows a non-monotonic pattern with SHORT performing worse than NO-COT. *(Right)*Accuracy plotted against mean actual completion tokens, showing that the LONG and UNCONSTRAINED conditions produce similar token counts despite different allocated budgets.

Table 4: Mean actual completion tokens by dataset and budget condition. Despite large differences in allocated budgets (150–3000), actual usage converges for the LONG and UNCONSTRAINED conditions.

Dataset	NO-COT	SHORT	MEDIUM	LONG	UNCONSTRAINED
MATH	116	163	383	537	610
GSM8K	73	76	148	186	183
MMLU-PRO	17	86	293	415	393
MUSR	23	78	327	423	399

MATH–GSM8K gap. This gap is negative (GSM8K outperforms MATH) and narrows with longer traces, from -0.46 at NO-COT to -0.16 at UNCONSTRAINED, reflecting MATH accuracy catching up as reasoning effort increases.

MATH–MMLU-Pro gap. Follows a similar narrowing pattern (-0.16 at NO-COT to -0.01 at UNCONSTRAINED), with longer traces closing the gap almost entirely.

MATH–MuSR gap. Exhibits a non-monotonic pattern. The gap *widens* from 0.36 (NO-COT) to 0.57 (SHORT) as short reasoning hurts MuSR more than MATH. It then narrows to 0.50 (LONG) before widening again to 0.57 (UNCONSTRAINED). The gap is minimized at the LONG condition, partially supporting our hypothesis that moderate-length reasoning optimizes robustness.

4.4 Confidence Calibration

Figure 3 presents confidence calibration results. We find a stark divergence between ID and OOD calibration quality.

On MATH, confidence is a strong predictor of correctness (ρ=0.633), making it a viable signal for adaptive reasoning depth. On GSM8K and MMLU-PRO, the correlation is moderate ($\rho\approx$0.30). On MUSR, however, the correlation is essentially zero ($\rho = -0.017$, p=0.74): the model reports high confidence ($> 75\%$) on nearly all questions despite achieving only 5–22% accuracy. This systematic overconfidence on hard OOD tasks represents a fundamental failure mode for any confidence-based adaptive strategy.

4.5 Adaptive Strategy Evaluation

The adaptive two-pass strategy (escalate from SHORT to LONG if confidence $< 75\%$) shows a clear split:

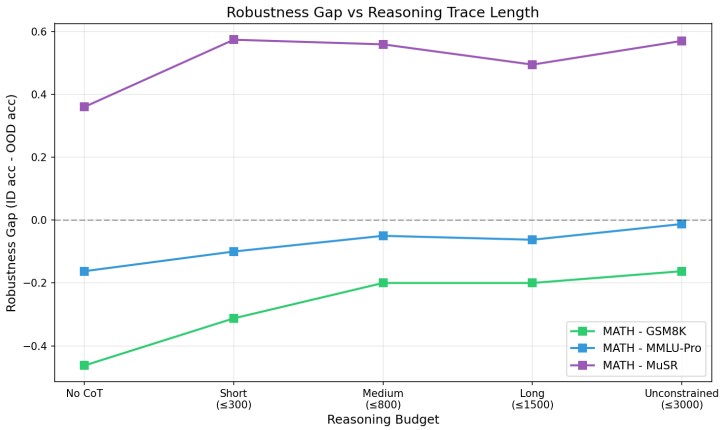


Figure 2: Robustness gap ($\text{Acc}_{\text{MATH}} - \text{Acc}_{\text{OOD}}$) by budget condition. The MATH–GSM8K gap is negative because GSM8K accuracy exceeds MATH. The MATH–MuSR gap is non-monotonic, widening at SHORT before narrowing at LONG.

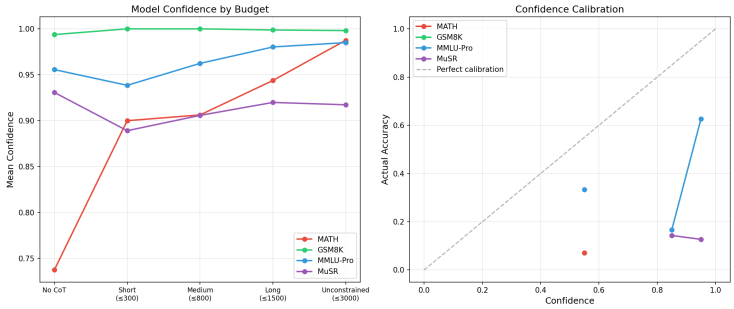


Figure 3: Confidence calibration by dataset. Spearman ρ between model-expressed confidence and correctness. Confidence is a strong signal on MATH (ρ=0.633, p<0.0001) but completely uninformative on MUSR ($\rho = -0.017$, p=0.74).

Success on MATH. On MATH, the adaptive strategy achieves 71.2% accuracy (matching LONG) while using only 366 mean tokens versus 537 for LONG—a 32% token reduction. This demonstrates that confidence-based escalation is viable in-distribution.

Failure on MuSR. On MUSR, the adaptive strategy achieves only 7.7%—worse than all fixed budgets at MEDIUM or above. The failure mode is clear: only 1 of 78 questions triggers escalation because the model is overconfident despite being wrong. The strategy defaults to SHORT answers, which we have shown are the worst-performing condition on this benchmark.

4.6 Token Efficiency

Figure 4 shows accuracy per token across conditions.

On GSM8K, NO-COT achieves the highest token efficiency because near-ceiling accuracy is reached with minimal tokens. On MATH, the SHORT condition is most efficient (balancing accuracy gain against token cost), though absolute accuracy continues to improve with longer traces. On MUSR, the LONG condition is most efficient, achieving the best balance of accuracy and token cost. These results suggest that optimal token allocation depends on both the task and the desired accuracy–efficiency trade-off.

4.7 Difficulty Interaction

Figure 5 shows the interaction between MATH difficulty level and budget condition.

Table 5: Confidence–correctness Spearman correlations by dataset. Model confidence is well-calibrated in-distribution but catastrophically miscalibrated on the hardest OOD benchmark.

Dataset	**Spearman** ρ	p**-value**
MATH	0.633	< 0.0001
GSM8K	0.304	< 0.0001
MMLU-PRO	0.301	< 0.0001
MUSR	-0.017	0.74

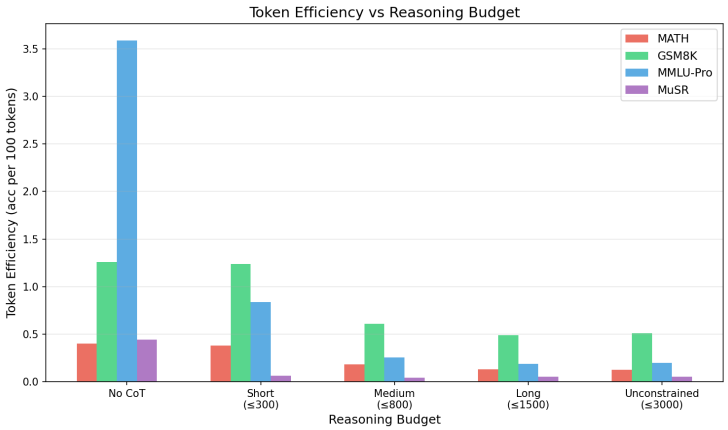


Figure 4: Token efficiency (accuracy / mean tokens $\times$ 100) by dataset and budget condition. On easy tasks (GSM8K), NO-COT is by far the most efficient. On harder tasks, efficiency peaks at moderate budgets before declining.

The benefit of longer reasoning is concentrated on harder problems. For Level 1–2 problems, SHORT achieves near-ceiling accuracy, while Level 5 problems show continued improvement through the UNCONSTRAINED condition. This difficulty-dependent effect is consistent with the task-level patterns we observe: easy tasks saturate quickly, while hard tasks benefit from extended reasoning up to a point.

5 Discussion

5.1 The Length–Accuracy Relationship Is Task-Dependent

Our central finding is that no single description captures the length–accuracy relationship across tasks. We observe three distinct regimes:

Within-capability tasks (MATH, MMLU-Pro). Accuracy increases monotonically with trace length, with diminishing returns. This aligns with the standard assumption that more reasoning helps on tasks the model can solve in principle. The Spearman correlations between budget level and accuracy are positive and significant (ρ=0.21, p<0.001 for MATH; ρ=0.12, p=0.02 for MMLU-PRO).

Easy tasks (GSM8K). Accuracy is invariant to trace length, with all conditions achieving $90-94\%$. Extended reasoning is wasted computation on tasks well within the model's capability.

Frontier tasks (MuSR). The relationship is non-monotonic. Short reasoning is worse than no reasoning (the "underthinking trap"), accuracy peaks at moderate length, and additional tokens provide marginal or negative returns. This pattern matches the error accumulation theory of Wu et al. [2025]: on problems at the model's capability frontier, additional reasoning steps introduce more noise than signal.

This task-dependent view reconciles apparent contradictions in prior work. Wu et al. [2025] observe the inverted U on in-distribution math tasks, which we confirm on our hardest OOD benchmark. Su et al. [2025] find that incorrect responses tend to be longer, consistent with overthinking on easy

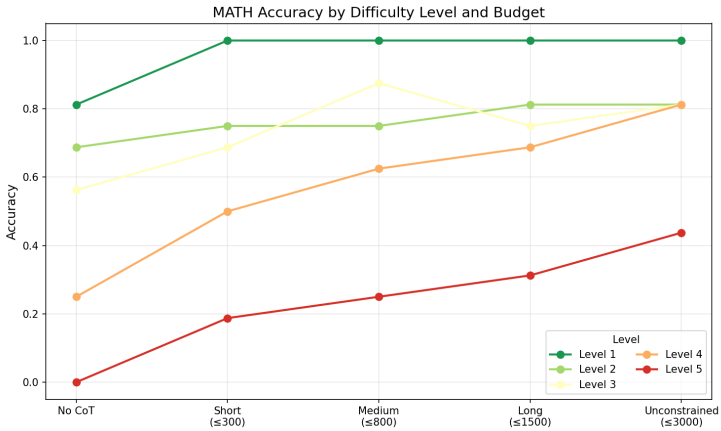


Figure 5: Accuracy on MATH by difficulty level and budget condition. Harder problems (Level 5) show the largest absolute benefit from longer reasoning, while easier problems (Level 1–2) see minimal improvement beyond SHORT.

problems and error accumulation on hard ones. The key insight is that the position on this spectrum depends on the interaction between task difficulty and model capability, not on absolute trace length alone.

5.2 The Underthinking Trap

The finding that SHORT (5.1%) underperforms NO-COT (10.3%) on MUSR is, to our knowledge, novel. We hypothesize two mechanisms. First, brief reasoning may *commit* the model to a reasoning direction without sufficient tokens to explore alternatives or self-correct—a form of premature closure. Second, the NO-COT condition may leverage the model's zero-shot pattern-matching capabilities, which can sometimes outperform shallow deliberation. This finding has practical implications: if forced to use short CoT (due to latency or cost constraints), practitioners may be better served by no CoT at all on sufficiently hard tasks.

This connects to Valmeekam and Kambhampati [2025], who show that trace content may be decoupled from reasoning quality. On frontier tasks, a few reasoning tokens may do more harm than good by steering the model toward a specific (and likely incorrect) inference path, whereas the no-CoT response draws on the model's full implicit knowledge without this anchoring effect.

5.3 Confidence Miscalibration as a Fundamental Barrier

The stark contrast in confidence calibration between MATH (ρ=0.633) and MUSR ($\rho\approx0$) represents a fundamental barrier to confidence-based adaptive reasoning systems. On MUSR, GPT-4.1 reports $> 75\%$ confidence on nearly all questions despite achieving only 5–22% accuracy. This is not merely weak calibration—it is *systematic overconfidence* on the hardest OOD tasks.

This miscalibration pattern is consistent with Dellibarda Varela et al. [2025], who show that token usage reflects self-assessed tractability rather than actual capability. The model "does not know what it does not know" on genuinely novel tasks. This poses a challenge for all adaptive systems that rely on internal uncertainty signals: they will fail precisely on the tasks where adaptation is most valuable.

Implications for adaptive systems. Our adaptive strategy achieves genuine token savings in-distribution (32% reduction on MATH) but fails catastrophically OOD. Future adaptive systems need external calibration signals—such as probe-based uncertainty estimation [Bogdan et al., 2025], semantic diversity of sampled reasoning paths, or learned difficulty classifiers—rather than relying on model self-reported confidence.

5.4 Self-Regulation of Output Length

An unexpected finding is that actual token usage converges between the LONG (max 1500) and UNCONSTRAINED (max 3000) conditions ($\sim$400$-$600 tokens across benchmarks). The model appears to reach a natural "saturation point" of reasoning depth regardless of the allocated budget ceiling. This suggests that prompt-based budget forcing primarily constrains *short* reasoning (preventing the model from reaching its natural depth) rather than encouraging *longer* reasoning beyond what the model would naturally produce. This has implications for budget-forcing studies: the effective range of prompt-based control is narrower than the nominal token limits suggest.

5.5 Limitations

Single model. We evaluate only GPT-4.1. The specific accuracy levels and optimal trace lengths likely differ across model families (e.g., DeepSeek-R1, Claude, open-source reasoning models). However, we expect the qualitative finding—that the length–accuracy relationship is task-dependent—to generalize.

Prompt-based budget forcing. System-prompt instructions provide imprecise control over actual reasoning length. RL-trained approaches such as L1 [Aggarwal and Welleck, 2025] achieve tighter length targeting but require model training. Our approach has the advantage of model-agnostic applicability.

Sample size. With 80 questions per dataset, some pairwise comparisons lack statistical power. We report bootstrap CIs throughout to characterize uncertainty, and note that the MUSR results (n=78) have the widest intervals due to both small sample size and low absolute accuracy.

Single run. Using T=0.0 ensures determinism within a run but prevents measuring inter-run variability. Stochastic decoding with multiple samples would provide richer distributional information.

Length vs. structure. We measure only trace length, not structural features such as branching, backtracking, or verification steps. Prior work [Jiang et al., 2025] suggests these features are more predictive of correctness than length alone. Our study establishes the baseline relationship that structural analyses can build upon.

No MuSR difficulty decomposition. MUSR contains three subtasks (murder mysteries, object placements, team allocation) that likely differ in difficulty and length sensitivity. Our balanced sampling across subtasks treats them as a single distribution.

6 Conclusion

We present a systematic investigation of reasoning trace length as a proxy for robustness across in-distribution and out-of-distribution benchmarks. Our experiments with GPT-4.1 across five budget conditions and four benchmarks reveal that the length–accuracy relationship is fundamentally task-dependent: monotonically increasing for tasks within the model's capability range, flat for easy tasks, and non-monotonic for tasks at the model's capability frontier. We identify the "underthinking trap"—a regime where short CoT is worse than no CoT—and demonstrate that model confidence is catastrophically miscalibrated on hard OOD tasks, rendering confidence-based adaptive strategies unreliable precisely where they are most needed.

The key takeaway for practitioners is that the "more reasoning is better" heuristic is insufficient. For in-distribution and moderate-difficulty tasks, longer reasoning generally helps with diminishing returns beyond $\sim$500 tokens. For tasks at the frontier of model capability, moderate-length reasoning outperforms both extremes, and adaptive strategies require external calibration signals rather than model self-reported confidence.

Future work should extend this analysis across multiple model families, employ RL-trained length control [Aggarwal and Welleck, 2025] for more precise budget targeting, and develop external uncertainty signals—such as probe-based methods [Bogdan et al., 2025], semantic entropy of sampled paths, or learned difficulty classifiers—that remain calibrated under distribution shift. Understanding the mechanistic basis of the underthinking trap and its interaction with model architecture is another promising direction.

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